objects undergo quantum jitter, for if they didn't we would know where they were and how fast they were moving with complete precision, in violation of Heisenberg's dictum. This holds true for the loops in string theory as well; no matter how placid a string appears it will always experience some amount of quantum vibration. The remarkable thing, as originally worked out in the 1970s, is that there can be energy cancellations between these quantum jitters and the more intuitive kind of string vibrations discussed above and illustrated in Figures 6.2 and 6.3. In effect, through the weirdness of quantum mechanics, the energy associated with the quantum jitters of a string is negative, and this reduces the overall energy content of a vibrating string by an amount that is roughly equal to Planck energy. This means that the lowest-energy vibrational string patterns, whose energies we would naively expect to be about equal to the Planck energy (i.e., 1 times the Planck energy), are largely canceled, thereby yielding relatively low net-energy vibrations—energies whose corresponding mass-equivalents are in the neighborhood of the matter and force particle masses shown in Tables 1.1 and 1.2. It is these lowest energy vibrational patterns, therefore, that should provide contact between the theoretical description of strings and the experimentally accessible world of particle physics. As an important example, Scherk and Schwarz found that for the vibrational pattern whose properties make it a candidate for the graviton messenger particle, the energy cancellations are perfect, resulting in a zero-mass gravitational-force particle. This is precisely what is expected for the graviton; the gravitational force is transmitted at light speed and only massless particles travel at this maximal velocity. But low-energy vibrational combinations are very much the exception rather than the rule. The more typical vibrating fundamental string corresponds to a particle whose mass is billions upon billions times greater than that of the proton.

This tells us that the comparatively light fundamental particles of Tables 1.1 and 1.2 should arise, in a sense, from the fine mist above the roaring ocean of energetic strings. Even a particle as heavy as the top quark, with a mass about 189 times that of the proton, can arise from a vibrating string only if the string's enormous characteristic Planck-scale energy is canceled by the jitters of quantum uncertainty to better